



Clinical Charge Guard

Developer-Ready Module & Feature Specification

Doctor Insurance Compliance, Procedure Bundling, Medical Necessity & Charge Audit

Recommended Product Name

Quilvian Clinical Charge Guard (CCG). Technical engine: Doctor Insurance Compliance & Charge Audit Engine (DICC AE). Position: submodule of Insurance Intelligence & Cashless Protection.

Metadata	Value
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Audience	Backend, Frontend, Fullstack, UI/Workflow Analyst, QA, DevOps, Product
Purpose	Implementation handoff for Codex-assisted development
Module Owner	Insurance Intelligence & Cashless Protection
Status	Developer Baseline
Date	12 August 2026

1. Naming, Positioning & Module Hierarchy

Nama yang direkomendasikan harus mudah dipahami dokter dan manajemen, tetapi tetap memiliki identifier teknis yang stabil untuk repository, API, database, dan event bus.

Layer	Recommended Name	Technical Identifier	Purpose
Product feature	Quilvian Clinical Charge Guard	clinical-charge-guard	Nama yang tampil di menu dan komunikasi pengguna.
Core engine	Doctor Insurance Compliance & Charge Audit Engine	doctor-insurance-compliance-engine	Mesin rule, compliance, alert, override, dan charge audit.
Feature	Procedure Bundling Guard	procedure-bundling	Mencegah unbundling, duplicate, incidental, dan mutually exclusive charge.
Feature	Medical Necessity Guard	medical-necessity	Memastikan tindakan tambahan memiliki indikasi, bukti, dan justifikasi.
Feature	Payer Rule Guard	payer-rule-engine	Menjalankan rule berbeda per payer/product/contract/effective date.
Feature	Charge Integrity Audit	charge-integrity	Mencocokkan order, performed service, operative note, consumable, dan billing.
Feature	Controlled Clinical Override	clinical-override	Memberi jalur penyimpangan klinis yang terdokumentasi dan dapat diaudit.
Feature	Doctor Compliance Analytics	doctor-compliance	Score, peer comparison, trend, dan corrective action.

Menu Recommendation

Insurance Intelligence > Clinical Charge Guard. Hindari nama “Doctor Audit” pada menu utama agar penerimaan klinis lebih baik. Istilah audit tetap digunakan pada fungsi internal dan laporan.

2. Business Objective

- Mencegah charge tidak valid sebelum invoice final dan sebelum klaim dikirim.
- Menurunkan risiko komplain payer, claim deduction, pending, rejection, audit eksternal, dan penghentian cashless.

- Menjadikan kontrak payer dan clinical governance sebagai rule yang dapat dieksekusi sistem.
- Memastikan keputusan klinis dokter tetap dapat berjalan pada kondisi yang sah melalui controlled override.
- Menyediakan satu audit trail yang menjelaskan siapa melakukan apa, rule apa yang terpicu, alasan keputusan, dan dampak finansialnya.

3. Non-Goals / Guardrails

- Sistem tidak menggantikan keputusan klinis dokter.
- Sistem tidak otomatis menyimpulkan fraud. Sistem hanya memberi compliance/risk flag untuk review.
- Sistem tidak menghapus transaksi historis; koreksi menggunakan reversal dan corrected transaction.
- Sistem tidak menganggap semua secondary procedure tidak valid. Exception yang klinis dan kontraktual tetap dapat ditagihkan.
- Sistem tidak menggunakan rule payer yang belum memiliki owner, sumber, versi, dan effective date.

4. User Roles & Permissions

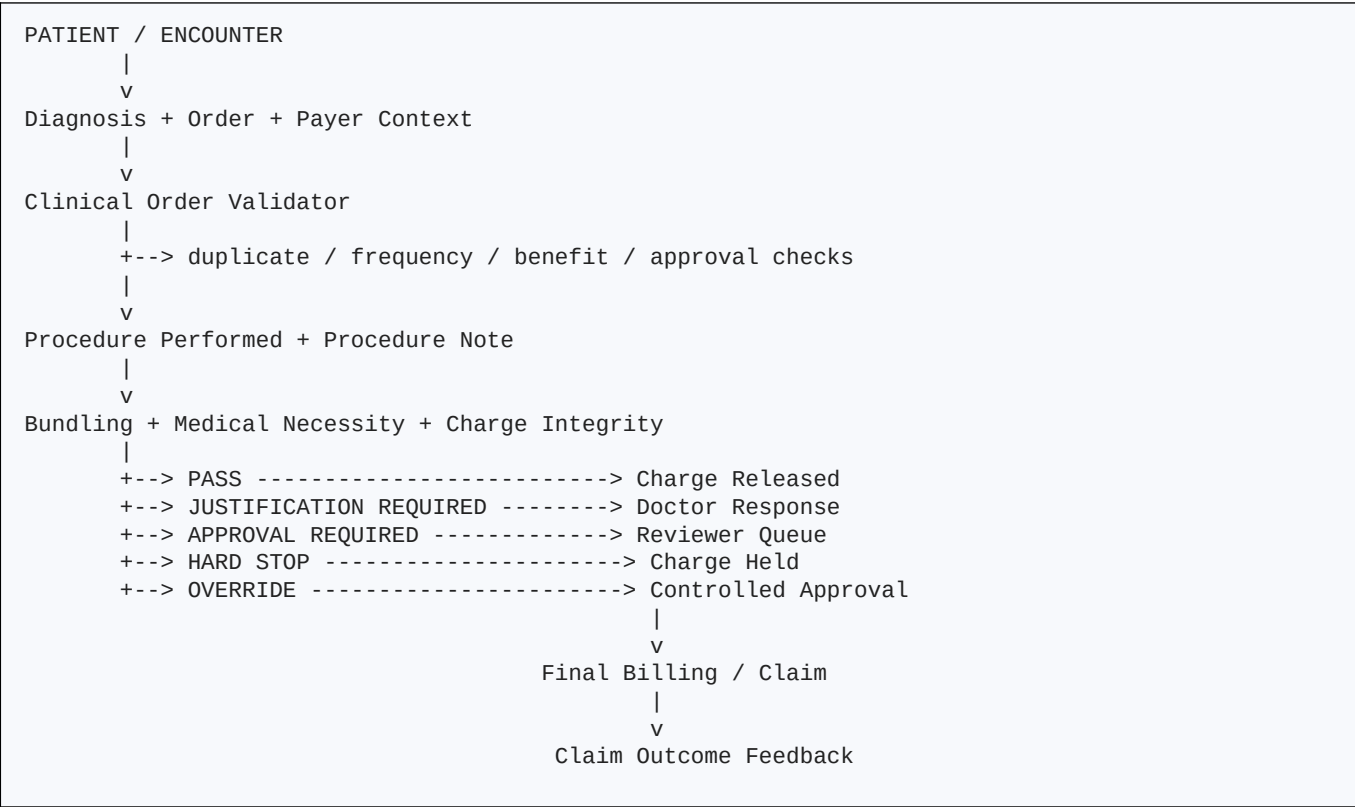
Role	Key Actions	Restricted Actions
Doctor/DPJP	Create order, view alert, add justification, request override.	Tidak mengubah payer rule/tarif; tidak approve override sendiri untuk high-risk case.
Operator/Procedure Doctor	Complete procedure note, identify primary/secondary procedure.	Tidak menghapus audit event.
Case Manager / Doctor LO	Review medical necessity, pathway deviation, emergency exception.	Tidak mengubah final contract rule.
Coder	Validate code, mapping, bundling, documentation sufficiency.	Tidak mengubah clinical note.
Insurance Officer	Maintain payer rule draft, approval/GL status, payer clarification.	Tidak publish rule tanpa maker-checker.
Billing	Review held charge, release/reverse after resolution.	Tidak bypass hard stop tanpa authorized override.
Medical Committee	Review recurrent patterns, peer review, clinical governance.	Tidak mengubah claim transaction directly.
Finance/Claim	Review financial impact, claim outcome, dispute.	Tidak edit clinical justification.
Admin/Rule Manager	Manage rule lifecycle, version, effective date.	Publish only with approval.

Role	Key Actions	Restricted Actions
Auditor	Read-only drilldown, export evidence.	No write/override.

5. Functional Components

ID	Component	Responsibility
CCG-01	Encounter Context Loader	Loads patient, encounter, payer, product, contract, benefit, limit, care setting.
CCG-02	Clinical Order Validator	Validates diagnosis, order, frequency, previous results, and required approval.
CCG-03	Procedure Bundling Engine	Determines primary/secondary relationship and bundling status.
CCG-04	Medical Necessity Engine	Validates independent indication and evidence sufficiency.
CCG-05	Procedure Note Reconciler	Matches operative/procedure note against coded procedures and charges.
CCG-06	Charge Integrity Engine	Detects duplicate, missing performed service, package conflict, mutually exclusive charges.
CCG-07	Payer Rule Engine	Selects rules by payer, product, contract, site, service date, and code version.
CCG-08	Alert & Decision Orchestrator	Produces soft alert, justification gate, hard stop, or override route.
CCG-09	Override Workflow	Routes authorization by risk, amount, urgency, and reason.
CCG-10	Audit Trail Service	Stores immutable event, previous/new values, evidence, decisions, and rule version.
CCG-11	Compliance Analytics	Doctor/unit/payer/rule metrics, peer comparison, trend, and drilldown.
CCG-12	Claim Feedback Loop	Maps payer outcome back to rule performance and case pattern.

6. End-to-End Workflow



6.1 Workflow Stage Details

Stage	Trigger	System Action	Output
1. Context Load	Encounter opened / payer selected	Load payer, product, contract, limit, benefit, active rule version.	Validation context
2. Doctor Order	Doctor saves order	Check duplicate, interval, benefit, approval, pathway, high-cost.	Pass / alert / hold
3. Procedure Documentation	Procedure completed / note signed	Extract performed procedure and evidence.	Procedure evidence set
4. Bundling Audit	Secondary procedure/code present	Classify included, separate, conditional, mutually exclusive, duplicate.	Bundling decision
5. Medical Necessity	Conditional secondary procedure	Check independent indication, findings, duration, complexity, approval.	Medical necessity decision

Stage	Trigger	System Action	Output
6. Charge Creation	Billing candidate created	Match order + performed service + documentation + payer rule.	Charge status
7. Exception Handling	Alert unresolved	Route clarification/approval/override.	Resolution record
8. Final Billing	All required controls resolved	Release compliant charges only.	Final bill
9. Claim Outcome	Payer response received	Map rejection/deduction/dispute reason to case/rule.	Feedback event

7. Core Business Rule Types

Rule	Type	Definition
R01	Diagnosis-to-Charge Separation	Diagnosis tidak boleh langsung membuat procedure charge. Wajib ada order/performed service/evidence.
R02	Duplicate Procedure/Charge	Same patient + encounter + procedure + time window + performer => duplicate candidate.
R03	Bundled / Included	Secondary step included in primary procedure/package => no separate charge by default.
R04	Mutually Exclusive	Dua code/tariff tidak boleh coexist pada kondisi payer tertentu.
R05	Incidental Procedure	Procedure minor/incidental to primary service => included unless exception criteria met.
R06	Add-on / Conditional	Procedure dapat ditagihkan hanya jika base procedure, condition, modifier, dan payer rule terpenuhi.
R07	Frequency / Interval	Repeat procedure/test harus memenuhi jumlah maksimum atau interval minimum.

Rule	Type	Definition
R08	Medical Necessity	Secondary procedure harus memiliki independent indication dan evidence klinis.
R09	Documentation Sufficiency	Charge tidak dapat final bila note wajib, site, approach, finding, atau outcome belum lengkap.
R10	Approval / GL Requirement	Procedure/item tertentu perlu approval valid sebelum financial release.
R11	Package Conflict	Komponen paket tidak boleh ditagih lagi secara individual.
R12	Performed Service Reconciliation	Charge harus memiliki bukti pelayanan dilakukan; planned-only/aborted perlu status khusus.
R13	High Cost Guard	Item/procedure di atas threshold memerlukan approval atau financial consent.
R14	Override Pattern	Override berulang melewati threshold menghasilkan compliance review.
R15	Contract Effective Date	Rule ditentukan berdasarkan service date dan versi kontrak aktif.

8. Alert Severity & Decision Model

Level	UI Behavior	When Used	Required Action
INFO	Banner/non-blocking	Informational payer/pathway note.	Acknowledge optional.
SOFT ALERT	Modal/banner; can continue	Low-risk repeat, recommendation, previous result available.	Doctor reviews and continues.
JUSTIFICATION REQUIRED	Blocking until reason entered	Deviation may be valid clinically.	Structured reason + optional narrative/evidence.
APPROVAL REQUIRED	Charge/order financial hold	Payer approval or reviewer decision needed.	Request approval / reviewer action.

Level	UI Behavior	When Used	Required Action
HARD STOP	Cannot release transaction	Definite duplicate, bundled charge, unsupported procedure, mutually exclusive conflict.	Correct transaction or authorized override if policy allows.
CRITICAL REVIEW	Escalated queue + management visibility	High-value/high-risk/recurrent pattern.	Dual approval / medical committee / insurance escalation.

9. Controlled Override

Override adalah mekanisme keselamatan agar sistem tidak menghambat kebutuhan klinis yang sah. Override bukan tombol bypass bebas.

Risk Tier	Example	Minimum Requirement	Approver
Low	Repeat low-cost test with changed condition	Reason code + note	Doctor self-justification
Medium	Secondary procedure with additional documented work	Evidence + structured justification	Case Manager / Doctor LO
High	High-cost procedure without prior payer approval	Evidence + financial exposure + urgency	Insurance + Medical reviewer
Critical	Large-value exception / recurrent high-risk behavior	Full case review + dual approval	Medical leadership + Finance/Insurance

- `override_reason_code`
- `override_reason_text`
- `clinical_evidence_refs`
- `risk_tier`
- `estimated_financial_exposure`
- `requested_by`
- `requested_at`
- `approved_by`
- `approved_at`
- `decision`
- `decision_note`
- `rule_id`
- `rule_version`

10. Reference Scenario - Appendectomy

Scenario

Diagnosis: acute appendicitis. Performed procedure: laparoscopic appendectomy. Proposed additional charges: laparoscopic adhesiolysis and laparoscopic debridement.

Procedure	Default Classification	Evidence Required for Separate Charge	Default Action
Laparoscopic appendectomy	PRIMARY / BILLABLE	Operative note confirms procedure.	Release if payer/benefit valid.
Laparoscopic adhesiolysis	BUNDLED / INCLUDED	Independent pathology/adhesion severity, distinct additional work, time/complexity, payer allows separate billing.	Hold/block unless exception criteria met.
Laparoscopic debridement	BUNDLED / INCLUDED	Independent necrotic/infected tissue requiring distinct work and payer allows separate billing.	Hold/block unless exception criteria met.

```

IF primary_procedure = LAP_APPENDectomy
AND secondary_procedure IN [LAP_ADHESIOLYSIS, LAP_DEBRIDEMENT]
AND payer_rule.relationship = INCLUDED
AND exception_criteria_met = FALSE
THEN charge_status = HOLD
    alert_severity = HARD_STOP
    reason = BUNDLED_WITH_PRIMARY_PROCEDURE
    
```

11. Rule Engine Specification

11.1 Rule Selection

Rule resolver harus memilih rule paling spesifik yang berlaku untuk encounter.

Rule key priority:

1. hospital_site + payer + product + contract + procedure + effective_date
2. hospital_site + payer + contract + procedure + effective_date
3. payer + procedure + effective_date
4. global procedure rule

Conflict resolution: more specific rule wins; if same specificity, latest approved version wins.

11.2 Rule Lifecycle

Status	Meaning	Allowed Actions
DRAFT	Rule being prepared.	Edit, simulate, submit.
IN_REVIEW	Awaiting maker-checker approval.	Approve/reject/comment.
APPROVED	Approved but not yet effective.	Schedule/publish.
ACTIVE	Used by engine.	Read, supersede.
SUSPENDED	Temporarily disabled with reason.	Resume/supersede.
RETIRED	No longer effective.	Read-only history.

11.3 Minimum Rule Fields

Field Group	Fields
Identity	rule_id, rule_code, name, description, rule_type, payer_id
Scope	product_id, contract_id, hospital_site_id, primary_procedure_code, secondary_procedure_code, diagnosis_scope, relationship
Decision	severity, action, exception_criteria, required_evidence, approval_required
Validity & Governance	effective_from, effective_to, priority, source_reference, owner_user_id, version, status

12. Domain Data Model

Entity	Minimum Fields
Encounter	encounter_id, patient_id, site_id, care_setting, admission_at, discharge_at, payer_context_id
PayerContext	payer_id, product_id, contract_id, policy_id, benefit_status, limit_status
ClinicalDiagnosis	diagnosis_id, encounter_id, code, description, type, onset, clinician_id

Entity	Minimum Fields
ClinicalOrder	order_id, encounter_id, order_code, ordered_by, ordered_at, status, clinical_indication
ProcedureRecord	procedure_id, encounter_id, procedure_code, approach, site, start_at, end_at, performer_id, status
ProcedureNote	note_id, procedure_id, signed_by, signed_at, findings, technique, complexity, complications
ChargeCandidate	charge_id, encounter_id, code, tariff_id, amount, source_type, source_id, status
ComplianceRule	rule_id, version, scope, relationship, severity, action, effective dates
ComplianceAlert	alert_id, encounter_id, rule_id, target_type, target_id, severity, status, evidence_snapshot
Justification	justification_id, alert_id, reason_code, text, evidence_refs, author_id, created_at
OverrideRequest	override_id, alert_id, risk_tier, financial_exposure, status, requester, approver
AuditEvent	event_id, actor_id, action, entity_type, entity_id, before_json, after_json, timestamp
ClaimOutcome	claim_id, encounter_id, payer_result, deduction_reason, amount_submitted, amount_paid

13. State Machines

13.1 Charge Candidate Status

```

DRAFT -> VALIDATING -> CLEARED -> RELEASED
      \-> HOLD_JUSTIFICATION -> CLEARED
      \-> HOLD_APPROVAL -----> CLEARED
      \-> BLOCKED -----> CORRECTED / CANCELLED
      \-> OVERRIDE_PENDING ----> CLEARED / BLOCKED

RELEASED -> REVERSED -> CORRECTED (if post-release correction required)
    
```

13.2 Alert Status

```

OPEN -> ACKNOWLEDGED -> JUSTIFIED -> RESOLVED
      \-> ESCALATED -> APPROVED_OVERRIDE -> RESOLVED
      \-> ESCALATED -> REJECTED_OVERRIDE -> BLOCKED
OPEN -> AUTO_RESOLVED (when source transaction corrected)
    
```

14. API Contract - Recommended

Nama endpoint bersifat baseline. Tim backend dapat menyesuaikan gateway/versioning, tetapi semantic contract sebaiknya dipertahankan.

Method	Endpoint	Purpose	Input	Output
POST	/v1/compliance/evaluate-order	Evaluate a new/updated clinical order.	order + encounter context	decision + alerts
POST	/v1/compliance/evaluate-procedure	Evaluate performed procedure and bundling.	procedure + note refs	classification + alerts
POST	/v1/compliance/evaluate-charge	Validate charge candidate before release.	charge + source refs	charge decision
GET	/v1/compliance/encounters/{id}/alerts	List current alerts.	encounter id	alerts
POST	/v1/compliance/alerts/{id}/justify	Submit clinical justification.	reason/evidence	updated decision
POST	/v1/compliance/alerts/{id}/override-requests	Create override request.	risk/reason/evidence	override id
POST	/v1/compliance/overrides/{id}/decision	Approve/reject override.	decision/note	updated status
POST	/v1/compliance/charges/{id}/release	Release cleared charge.	actor	released charge
POST	/v1/compliance/charges/{id}/reverse	Reverse released charge.	reason	reversal transaction
GET	/v1/compliance/rules	Search rules.	filters	rule list
POST	/v1/compliance/rules/simulate	Run rule simulation.	test context	matched rules/impact
POST	/v1/compliance/rules/{id}/publish	Publish approved rule.	version/effective date	active rule
POST	/v1/compliance/claim-outcomes	Ingest payer outcome.	claim result	feedback event

15. Event Model

Event	Consumer Behavior
clinical.order.created	Trigger order validation.
clinical.order.updated	Re-evaluate impacted alerts.
procedure.completed	Start procedure/medical necessity evaluation.
procedure.note.signed	Refresh evidence sufficiency.
charge.candidate.created	Run charge integrity validation.
charge.candidate.updated	Re-evaluate if amount/code/source changes.
payer.approval.updated	Resolve approval-dependent alerts.
compliance.alert.created	Notify responsible workflow.
compliance.override.requested	Create reviewer task.
compliance.override.decided	Update alert/charge status.
charge.released	Record release audit event.
claim.outcome.received	Update feedback analytics.

16. UI / Screen Specification

ID	Screen	Purpose
Screen 1	Clinical Charge Guard - Encounter Panel	Embedded side panel in EMR/billing showing payer, risk, active alerts, unresolved requirements.
Screen 2	Doctor Alert Dialog	Rule explanation, previous evidence, reason choices, continue/justify/request override actions.
Screen 3	Procedure Audit View	Primary/secondary procedures, bundling relationship, note evidence, payer rule, charge status.
Screen 4	Reviewer Work Queue	Filters by severity, payer, doctor, unit, amount, SLA; approve/reject/request info.

ID	Screen	Purpose
Screen 5	Charge Integrity Worklist	Held charges with cause, financial exposure, source reconciliation.
Screen 6	Payer Rule Manager	Rule create/edit/simulate/version/approve/publish.
Screen 7	Doctor Compliance Dashboard	Score, trend, top rule violations, peer benchmark, drilldown.
Screen 8	Executive Risk Dashboard	Prevented leakage, dispute trend, payer risk, high-risk doctors/units, unresolved critical cases.
Screen 9	Audit Evidence Viewer	Chronology of order, note, charge, alert, justification, override, claim outcome.

17. Doctor Alert UX Requirements

- Alert harus menjelaskan “apa yang terdeteksi”, “mengapa penting”, “dasar payer/contract”, dan “apa tindakan berikutnya”.
- Jangan menampilkan jargon internal seperti rule engine ID sebagai informasi utama; ID tersedia di detail teknis.
- Dokter dapat melihat evidence yang sudah ada agar tidak perlu mengulang dokumentasi.
- Untuk emergency, jangan memblokir tindakan klinis; financial/administrative hold berjalan paralel.
- Alert yang sama tidak boleh muncul berulang tanpa perubahan konteks (dedup/suppression).
- High-risk hard stop harus menyediakan jalur “Correct”, “Request Review”, dan bila diizinkan “Emergency Override”.

18. Doctor Compliance Analytics

Metric	Definition	Use
Bundling Violation Rate	Cases with bundling alert / eligible procedures.	Education and peer review.
Unsupported Secondary Procedure Rate	Secondary charge without sufficient evidence.	Documentation improvement.
Duplicate Order/Charge Rate	Confirmed duplicate / eligible transactions.	Process quality.
Approval Compliance	Required approval obtained before release.	Payer compliance.

Metric	Definition	Use
Override Rate	Overrides / high-severity alerts.	Pattern monitoring.
Override Acceptance Rate	Approved overrides / override requests.	Rule tuning and behavior review.
Claim Impact	Deduction/rejection amount linked to physician-related rule.	Financial feedback.
Documentation Completeness	Required evidence present at first review.	Clinical documentation quality.

Governance Note

Doctor Compliance Score tidak boleh menjadi satu-satunya dasar sanksi. Gunakan sebagai signal untuk peer review, edukasi, dan root-cause analysis dengan case-mix/context yang memadai.

19. Audit Trail Requirements

Setiap perubahan keputusan harus dapat direkonstruksi end-to-end.

- Actor: user_id, role, device/session if available.
- Action: create/update/delete-request/reverse/justify/approve/reject/release.
- Entity: encounter, order, procedure, charge, rule, alert, override, claim.
- Before/after snapshot untuk field kritis.
- Rule ID + version + effective date yang digunakan saat keputusan.
- Evidence snapshot/reference saat alert dibuat.
- Timestamp UTC plus display timezone.
- Correlation ID agar semua event satu episode dapat ditelusuri.
- Tidak boleh ada hard delete untuk audit-relevant entities.

20. Error, Downtime & Fallback Behavior

Condition	Required Behavior
Rule engine timeout	Do not silently pass. Mark validation_pending and queue retry; show clear status.
Payer rule unavailable	Use global rule only if configured; otherwise require manual review for high-risk charge.
EMR integration down	Capture deferred validation task; reconcile after recovery.
Billing integration down	Do not lose decision; persist release/hold state and replay.

Condition	Required Behavior
Approval service unavailable	Keep charge in HOLD_APPROVAL; clinical emergency pathway remains available.
Duplicate event delivery	Idempotency key prevents duplicate alert/decision.
Rule conflict	Escalate to rule governance; do not choose randomly.

21. Security & Privacy

- RBAC with least privilege; clinical notes only visible to authorized roles.
- Maker-checker for rule publication and critical configuration changes.
- Encryption in transit and at rest.
- Immutable audit logging with retention policy.
- Sensitive diagnoses and attachments require field-level authorization.
- Exports must be logged and optionally watermarked with user/time.
- No training/learning process may expose patient identifiers outside approved environment.

22. Non-Functional Requirements

NFR	Target
Synchronous rule response	P95 <= 2 seconds for order/charge checks.
Complex procedure evaluation	P95 <= 5 seconds.
Availability	>= 99.5% monthly; no silent bypass.
Idempotency	All mutation endpoints support idempotency key.
Traceability	100% high-severity decisions traceable to rule/evidence/user.
Scalability	Multi-hospital, multi-payer, multi-contract, versioned code sets.
Configurability	New rule/threshold without application redeploy after approval.
Observability	Structured logs, metrics, trace ID, rule hit rate, error rate, latency.
Localization	Indonesian user-facing terminology; technical identifiers remain English.
Accessibility	Keyboard-friendly dialogs, clear severity labels, readable contrast.

23. Acceptance Criteria / UAT

ID	Scenario	Expected Result
UAT-01	Appendectomy + routine adhesiolysis/debridement	Secondary charges are held as included unless exception evidence and payer allowance exist.
UAT-02	Valid independent secondary procedure	System routes to medical necessity/payer approval and can release after criteria satisfied.
UAT-03	Duplicate charge	Same service is blocked before final billing.
UAT-04	Procedure absent from signed note	Charge cannot be released.
UAT-05	Repeat test inside restricted interval	Justification required or blocked per payer rule.
UAT-06	High-cost item without approval	Charge is held; emergency clinical flow remains available.
UAT-07	Override high-risk case	Dual approval required; all evidence and decisions logged.
UAT-08	Rule effective date	Old encounter uses old rule; new service date uses new active version.
UAT-09	Correction after release	Original charge remains; reversal + corrected transaction created.
UAT-10	Payer-specific behavior	Same procedure can produce different decision for different payer/product contracts.
UAT-11	Downtime recovery	Queued validation reconciles without duplicate alerts.
UAT-12	Dashboard drilldown	Metric can drill to encounter -> order -> procedure -> charge -> rule -> outcome.

24. Suggested Development Epics

Epic	Name	Scope
EPIC-1	Foundation & Context	Encounter context, payer context, rule resolver, audit framework.
EPIC-2	Order Compliance	Duplicate/frequency/high-cost/approval checks in CPOE.
EPIC-3	Procedure Bundling	Primary-secondary classification and bundling rule evaluation.
EPIC-4	Medical Necessity	Structured evidence, justification, reviewer workflow.
EPIC-5	Charge Integrity	Reconciliation, hold/release/reversal, package conflict.
EPIC-6	Override Workflow	Risk tiers, approval matrix, notifications, SLA.
EPIC-7	Rule Management	Maker-checker, versioning, simulation, effective dates.
EPIC-8	Analytics	Doctor/payer/unit dashboards and drilldown.
EPIC-9	Claim Feedback	Outcome ingestion, deduction mapping, rule quality feedback.
EPIC-10	Hardening	Security, performance, observability, downtime reconciliation.

25. Suggested MVP Scope

MVP Recommendation

Jangan langsung membangun semua specialty. Mulai dari procedure dengan volume/financial risk tinggi dan rule payer yang paling jelas. Fokus membuktikan prevention sebelum memperluas intelligence.

- 5-10 payer utama.
- 10-20 procedure prioritas: appendectomy, cholecystectomy, hernia repair, common endoscopy, selected orthopedic and obstetric procedures sesuai volume rumah sakit.
- Rule types MVP: duplicate, bundled/included, mutually exclusive, approval required, documentation sufficiency, high-cost.

- UI MVP: encounter panel, doctor alert, reviewer queue, rule manager, charge integrity worklist.
- Analytics MVP: prevented charge value, alert count, override rate, claim deduction baseline.

26. Definition of Done

- Functional requirements implemented and mapped to test cases.
- All high-severity decisions have explainable rule + evidence.
- No silent pass during integration failure.
- Rule publication uses maker-checker and versioning.
- Charge status reconciles with billing source of truth.
- Audit trail reconstructs full case chronology.
- Role permissions validated.
- Performance target met under expected load.
- UAT signed by Medical, Insurance, Billing, Finance, and Product.
- Operational SOP and rollback plan available before production go-live.

27. Codex Development Handoff Prompt

Bagian ini dapat disalin sebagai prompt awal saat tim menggunakan Codex untuk memecah implementasi ke repository aktual. Developer tetap harus menambahkan struktur repo, tech stack, service names, database, dan coding conventions yang berlaku.

You are implementing the Quilvian Clinical Charge Guard module.

Primary goal:

- Prevent non-compliant clinical orders and charges before final billing/claim submission.
- Preserve physician clinical autonomy through controlled, auditable overrides.

Core capabilities:

1. Encounter and payer context resolution
2. Clinical order validation
3. Procedure bundling classification
4. Medical necessity validation
5. Charge integrity reconciliation
6. Payer-specific rule resolution by effective date
7. Alert/justification/approval/hard-stop orchestration
8. Controlled override workflow
9. Immutable audit trail
10. Compliance analytics and claim feedback

Implementation constraints:

- Do not hard-delete audit-relevant records.
- Use idempotency for mutation/event handlers.
- Never silently bypass a failed rule-engine call.
- Emergency clinical care can proceed; financial release may remain on hold.
- Every decision must reference rule_id, rule_version, and evidence snapshot.
- Separate clinical necessity from billability: a clinically necessary action can still be non-billable.

First task:

- Inspect the existing repository and identify services, tables, and UI areas that map to the entities and APIs in this specification.
- Produce an implementation plan by epic, schema changes, endpoints/events, migration risks, and test strategy before coding.
- Do not invent payer rules; use seeded/configured rules only.

28. Developer Checklist Before Coding

- [] Confirm source of truth for patient/encounter ID.
- [] Confirm source of truth for payer/product/contract.
- [] Confirm current order and billing lifecycle/statuses.
- [] Confirm procedure and charge coding dictionaries.
- [] Confirm how operative/procedure notes are stored and signed.
- [] Confirm approval/GL service and status model.
- [] Confirm existing audit logging/event bus infrastructure.
- [] Confirm identity/RBAC implementation.
- [] Confirm whether charge reversal exists and how it posts to finance.
- [] Confirm priority procedures and payer rules for pilot.
- [] Confirm emergency override policy and approvers.
- [] Confirm reporting/data warehouse integration.

29. Glossary

Term	Definition
Bundling	Beberapa komponen layanan diperlakukan sebagai satu paket/tindakan utama dan tidak ditagih terpisah.
Unbundling	Memecah komponen integral menjadi charge terpisah yang tidak dibenarkan oleh rule.
Primary Procedure	Tindakan utama yang merepresentasikan tujuan terapeutik utama episode/prosedur.
Secondary Procedure	Tindakan tambahan yang dapat included, conditional, atau separately billable.
Medical Necessity	Dasar klinis yang menunjukkan tindakan diperlukan untuk kondisi pasien.
Hard Stop	Kontrol yang mencegah transaksi dilanjutkan hingga dikoreksi atau dioverride secara sah.
Controlled Override	Penyimpangan dari rule melalui alasan, evidence, dan approval sesuai risiko.
Charge Integrity	Kesesuaian charge dengan order, pelayanan aktual, dokumentasi, paket, dan kontrak payer.
Payer Rule	Ketentuan billability/approval/bundling/frequency yang berlaku pada payer/product/contract tertentu.

Term	Definition
Evidence Snapshot	Salinan referensi data yang digunakan engine pada saat keputusan dibuat.

30. Final Module Summary

What the module does

Clinical Charge Guard menjadi control layer antara keputusan klinis dan financial charge. Ia tidak menentukan terapi, tetapi memeriksa apakah order dan charge dapat dipertanggungjawabkan secara klinis, terdokumentasi, tidak terduplikasi, sesuai bundling, sesuai approval, dan sesuai kontrak payer sebelum tagihan menjadi masalah.

Implementasi yang berhasil menghasilkan tiga keluaran utama: (1) pencegahan charge bermasalah sebelum invoice, (2) workflow exception yang aman bagi dokter dan pasien, dan (3) data compliance yang dapat digunakan untuk memperbaiki rule, hubungan payer, dan mutu dokumentasi klinis.